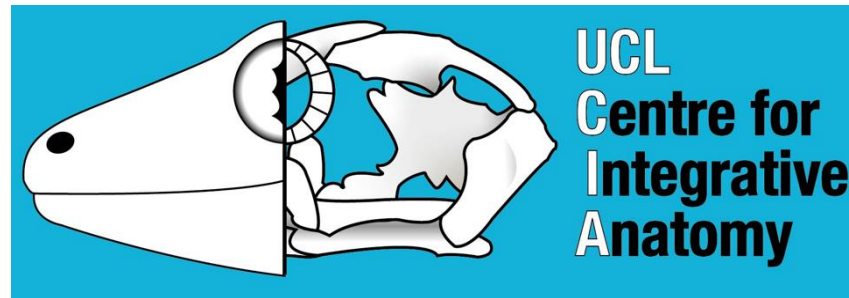
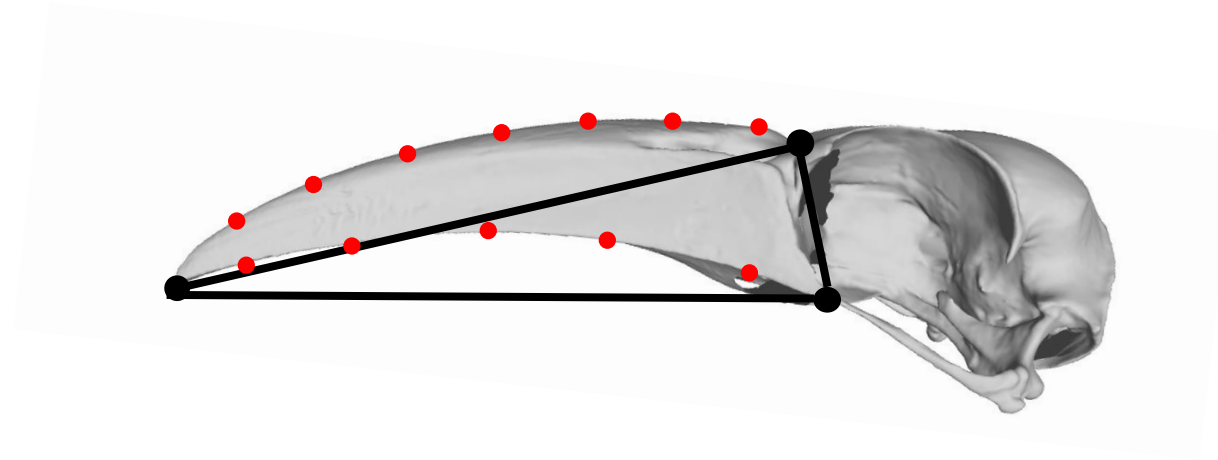
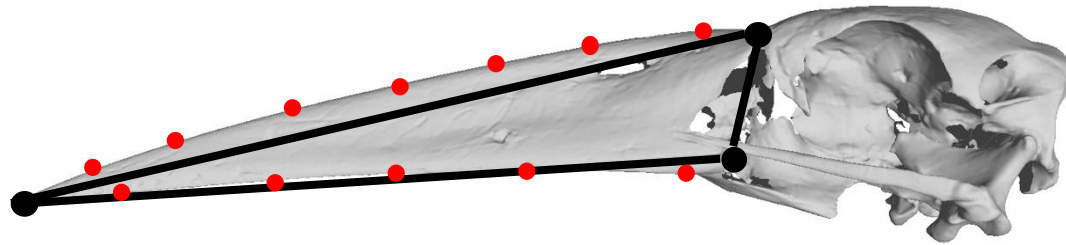


# Advanced Shape Analysis

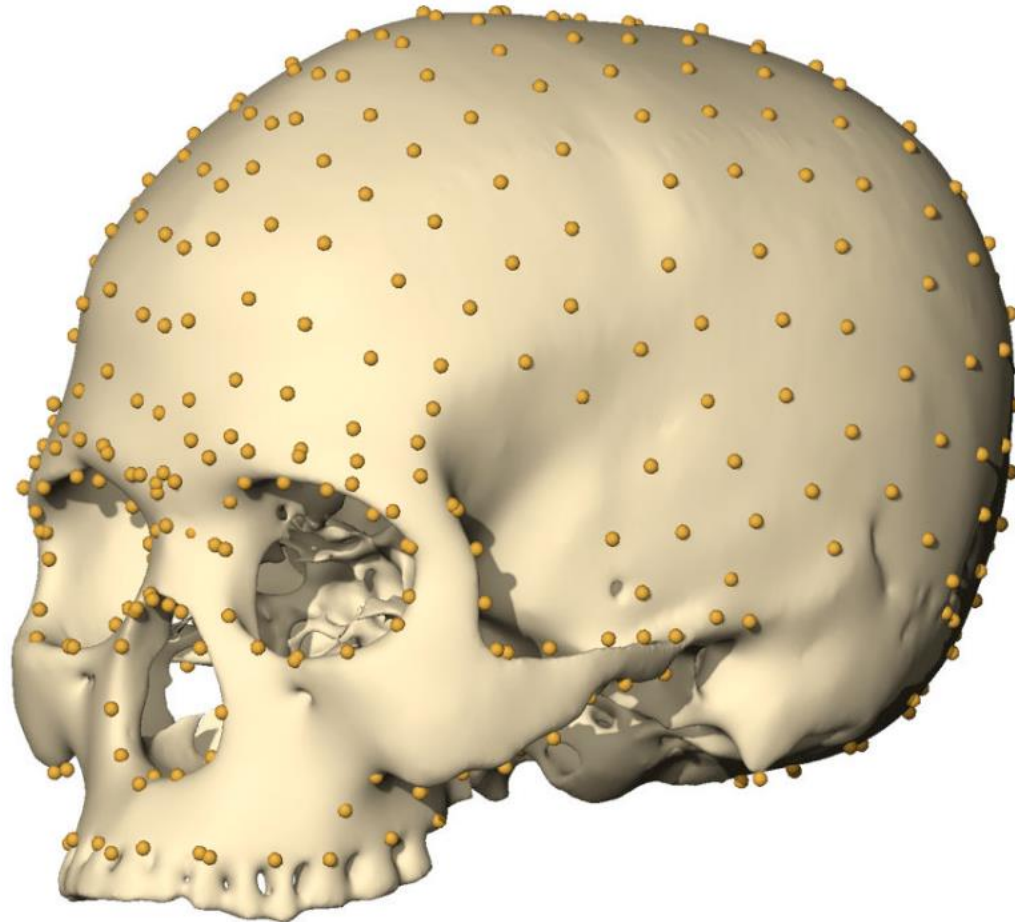
Ryan N. Felice



# Sliding Semilandmarks

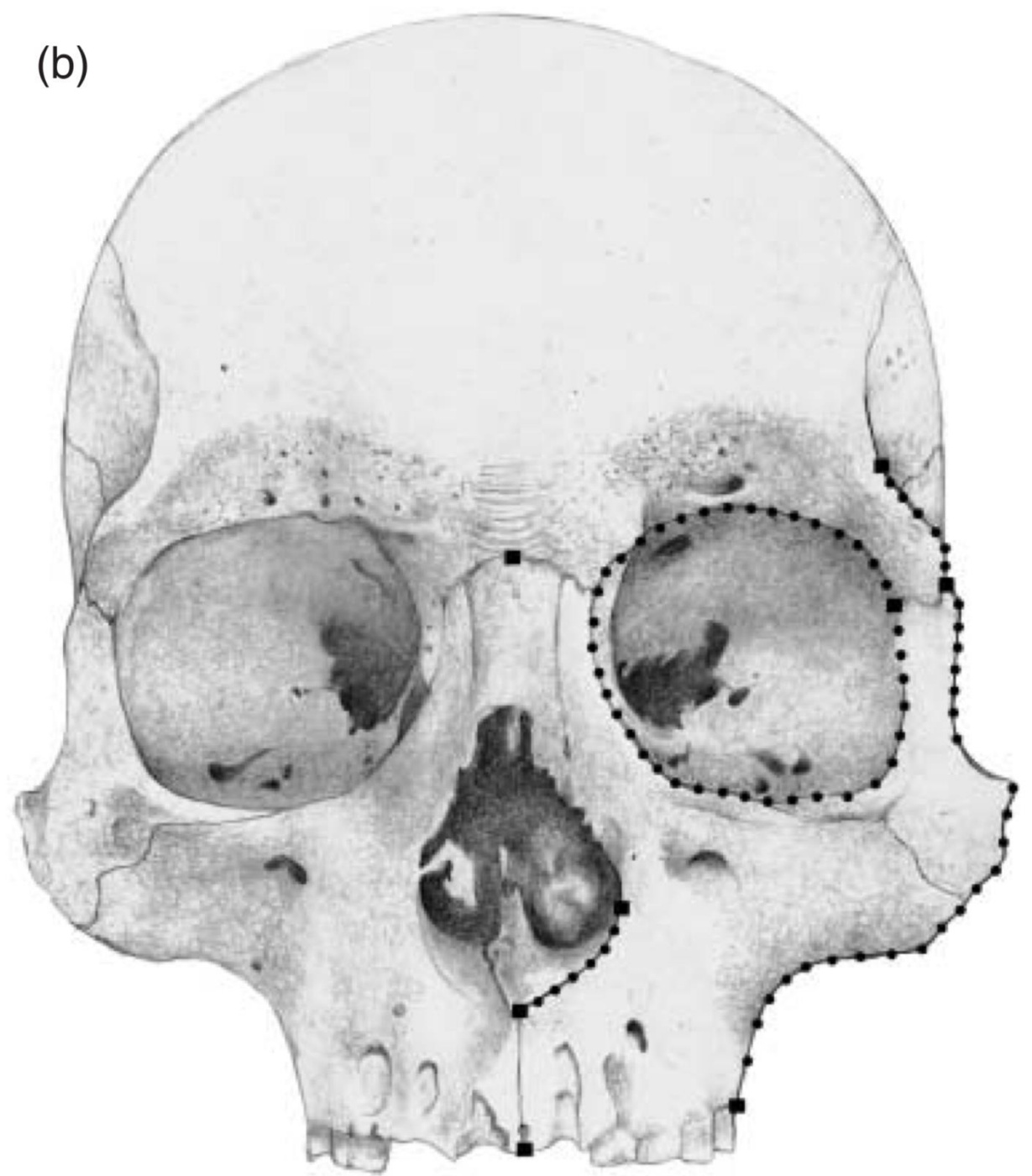


# Surface Semi-landmarks



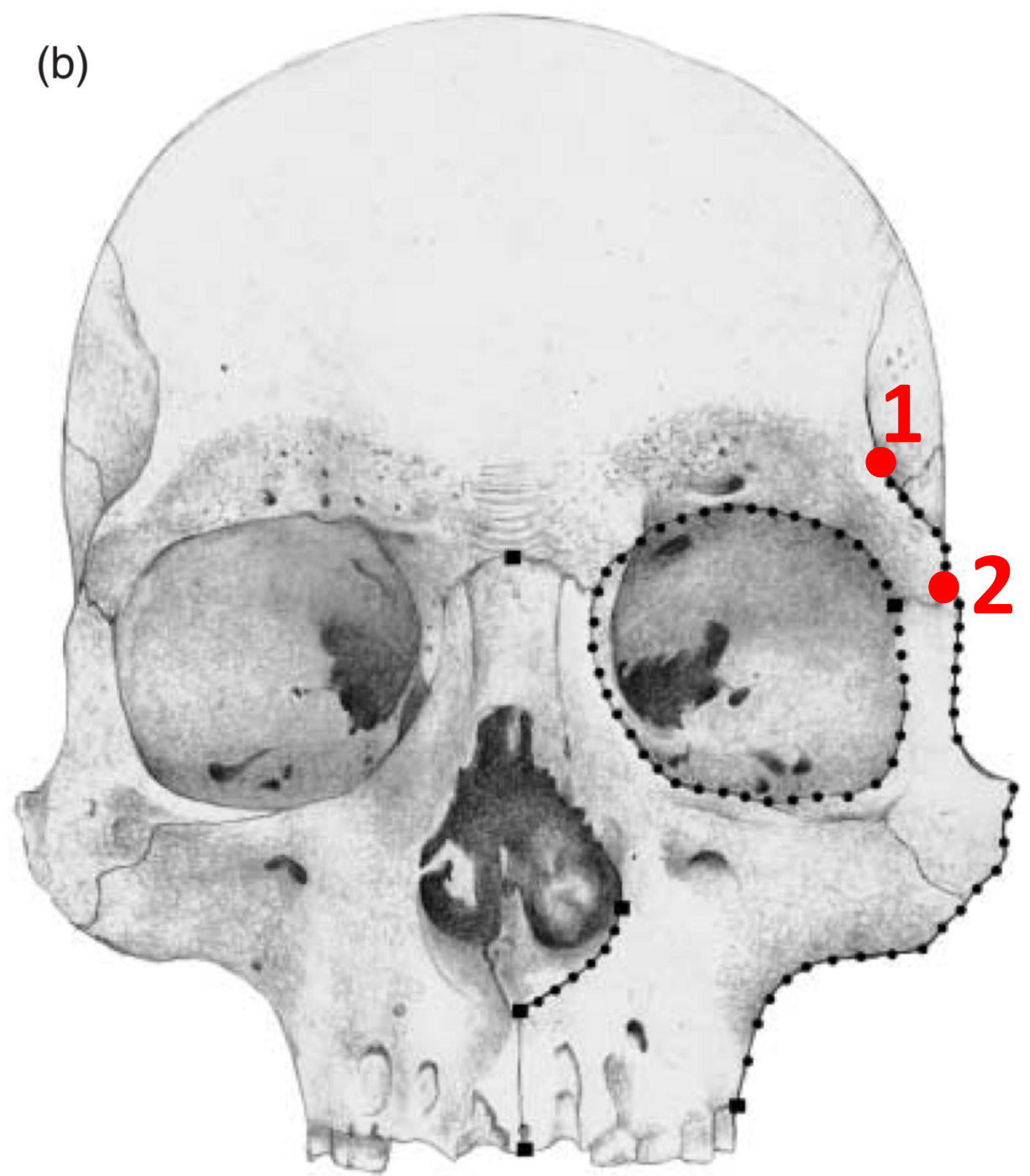
# Sliding Semilandmarks

(b)



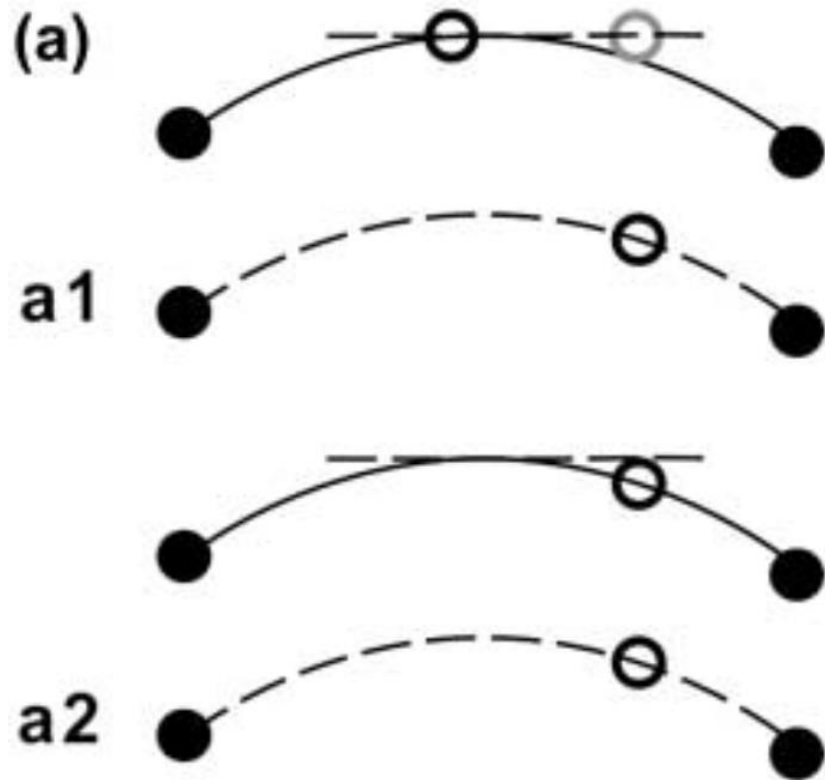
# Sliding Semilandmarks

(b)

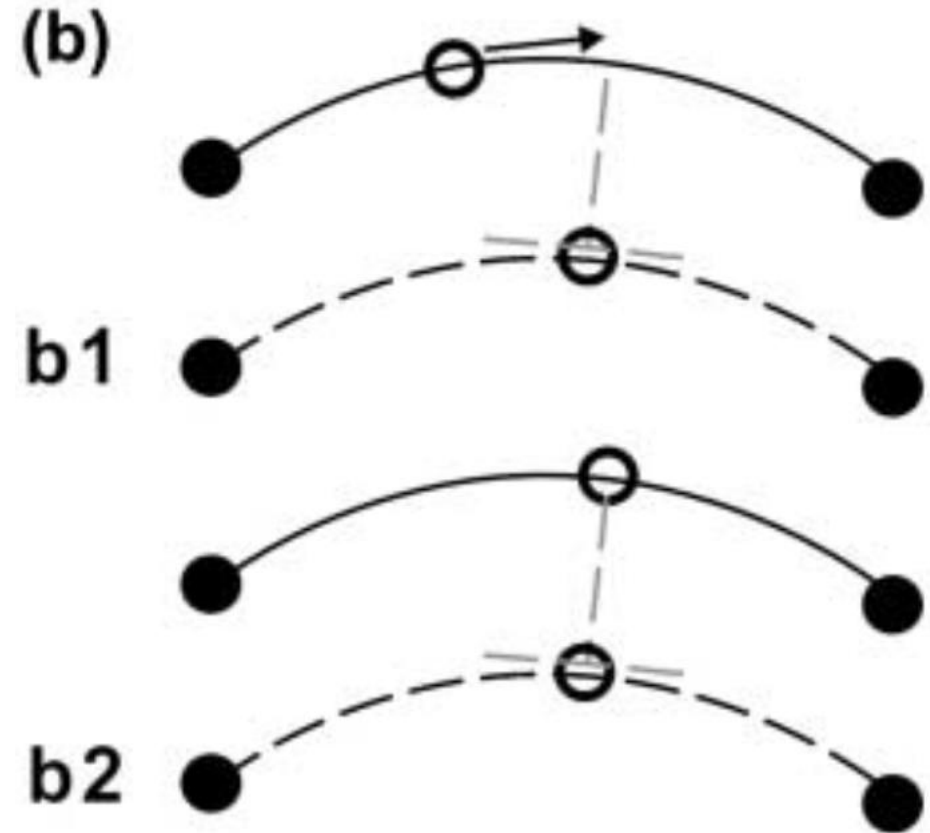


# Sliding Semilandmarks

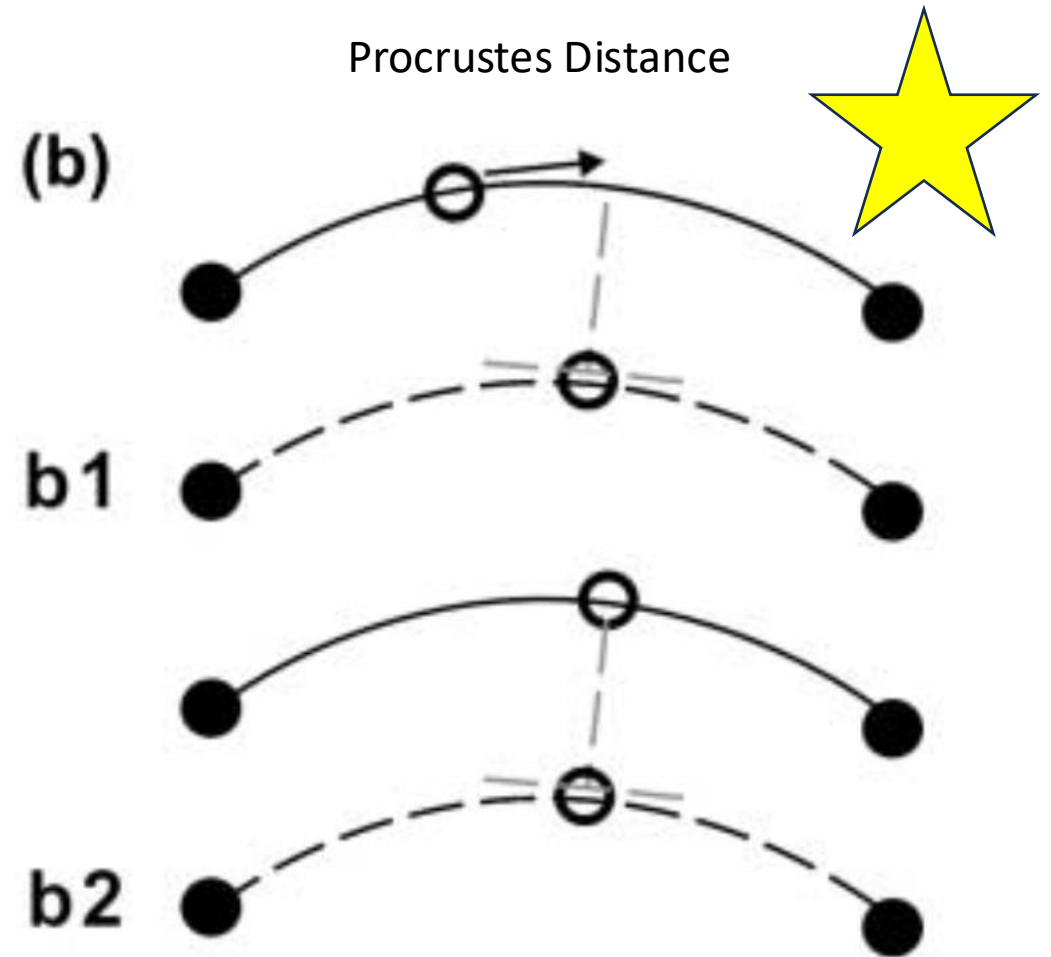
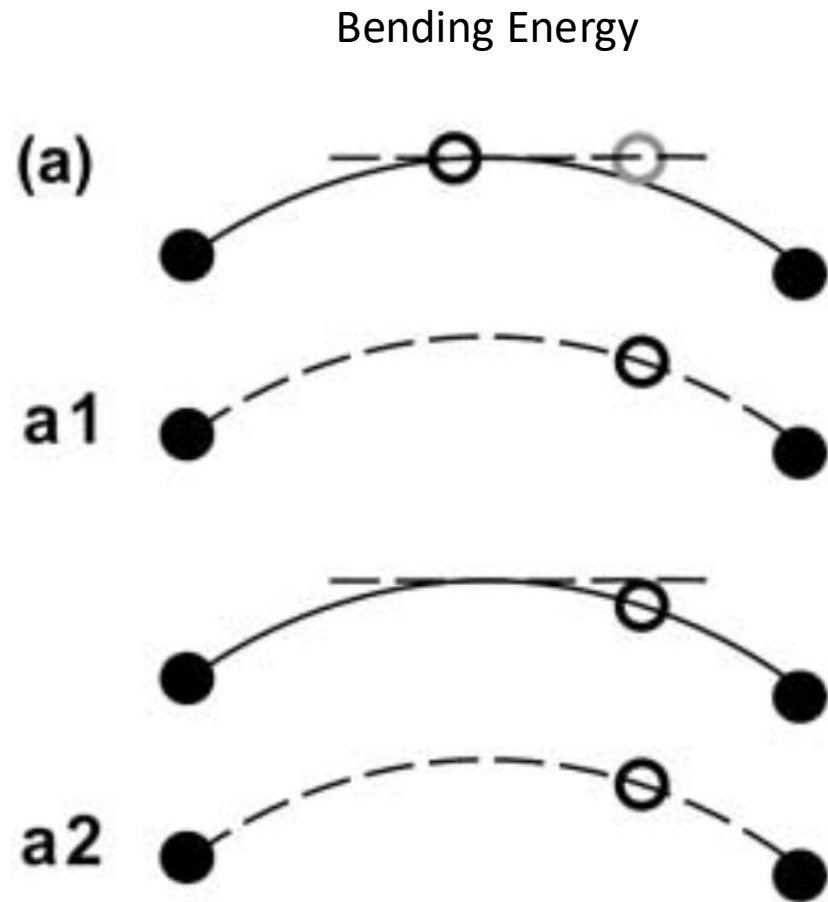
Bending Energy



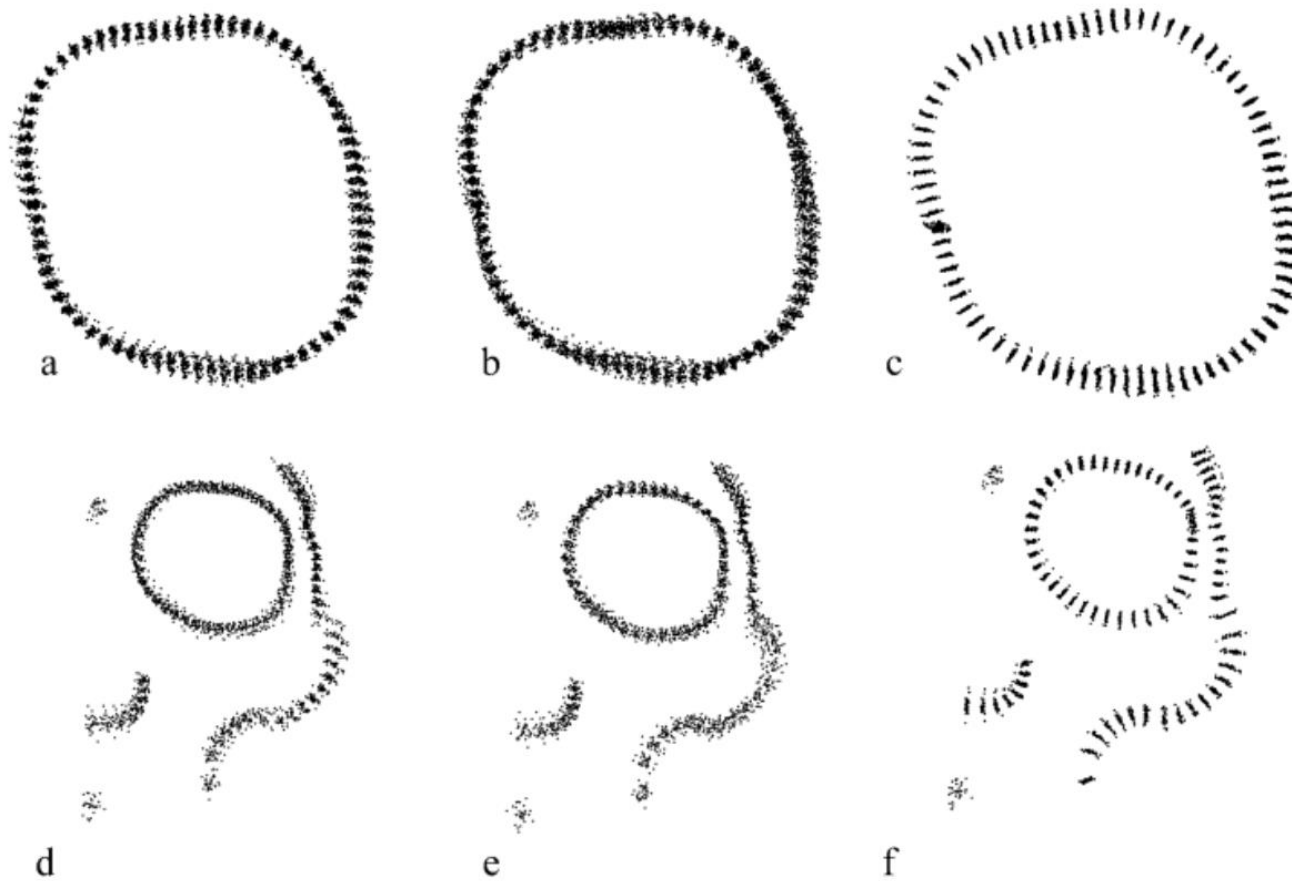
Procrustes Distance



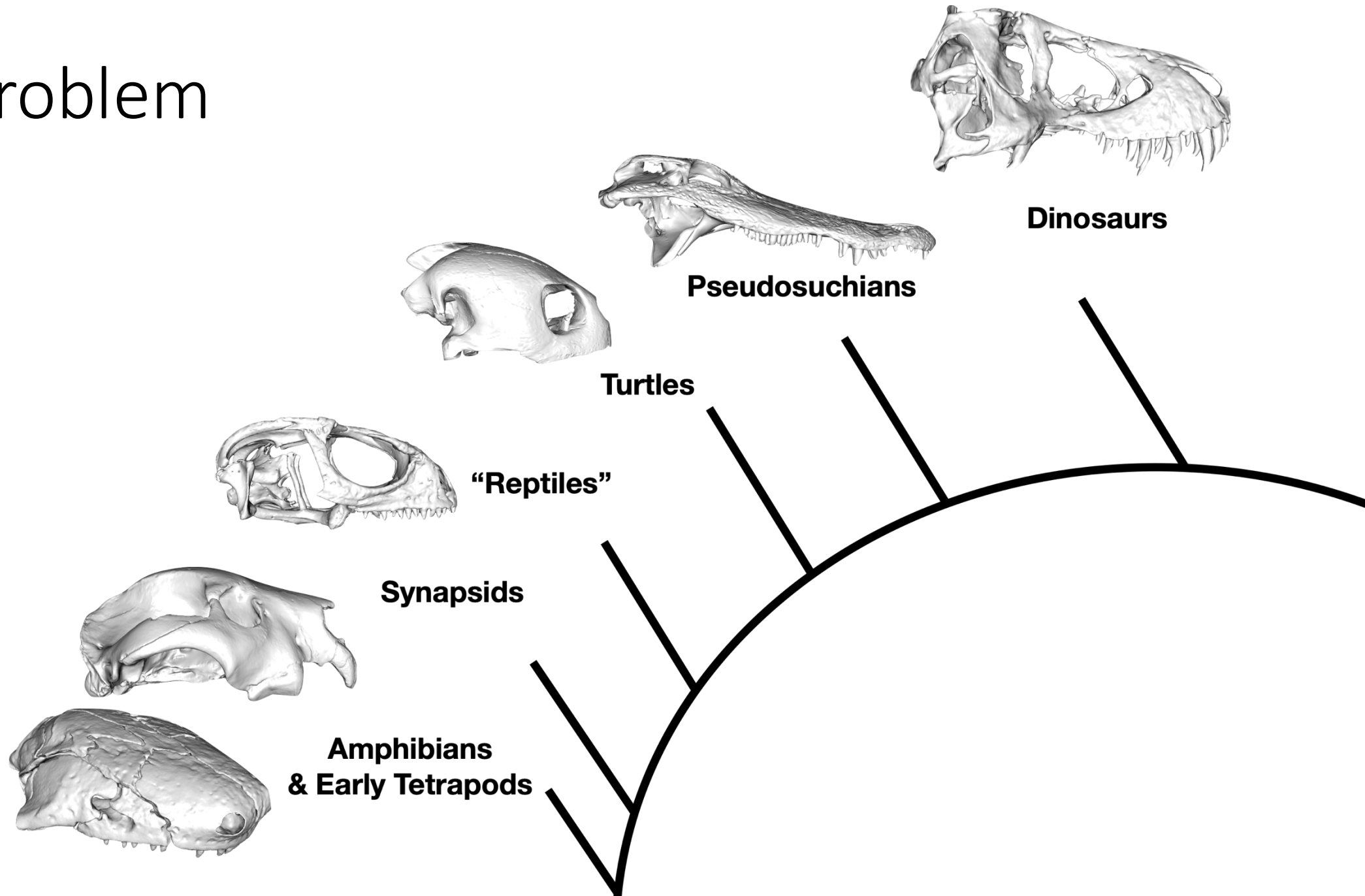
# Sliding Semilandmarks

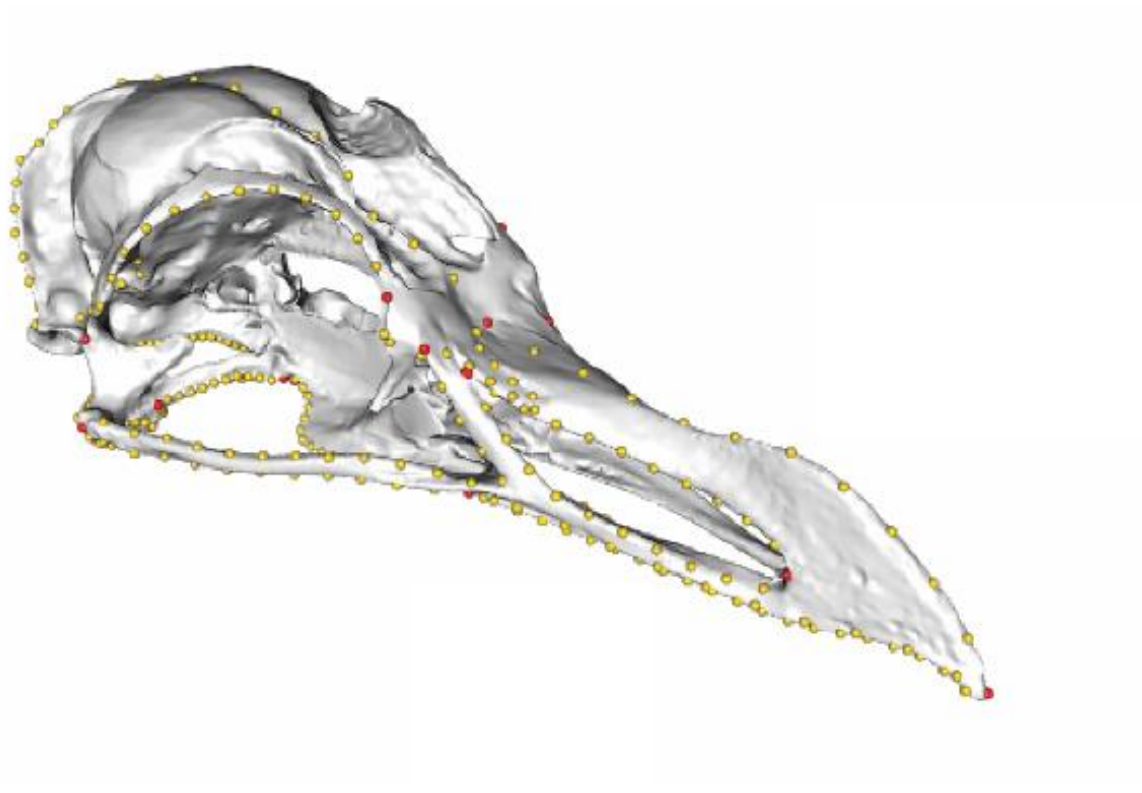


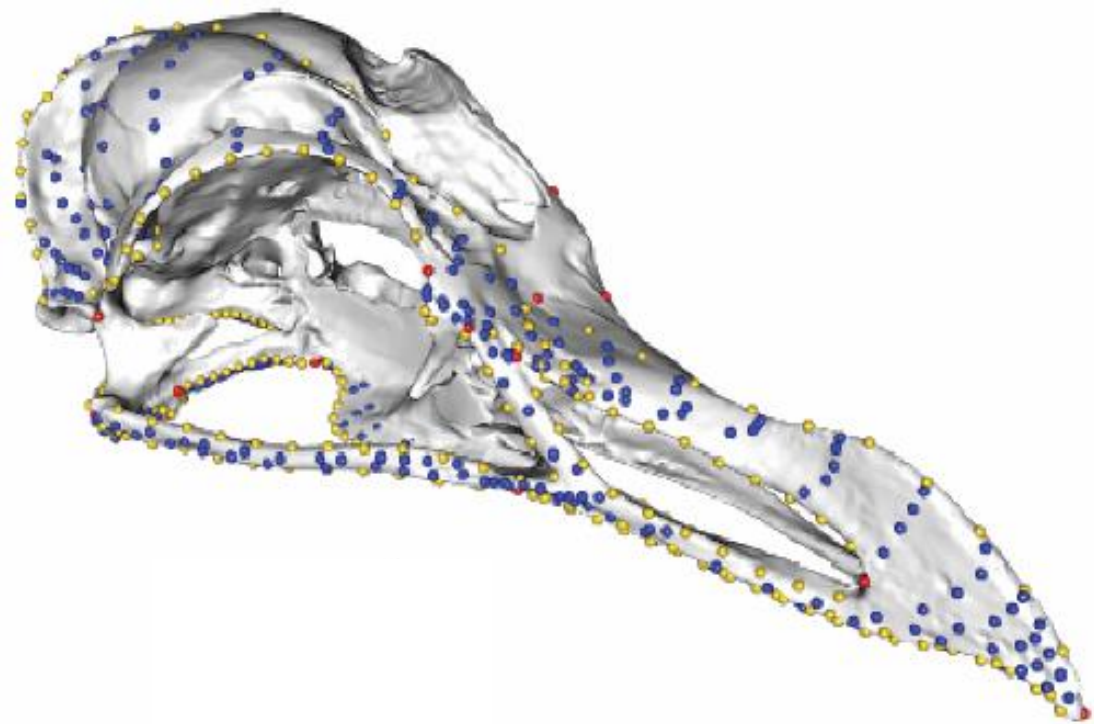
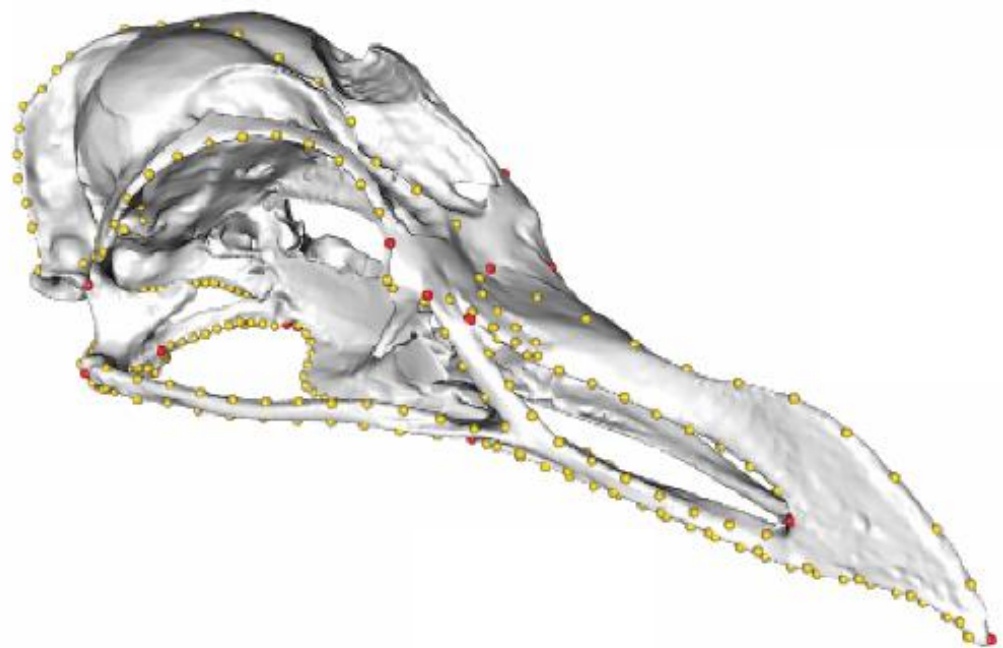
# Sliding Semilandmarks



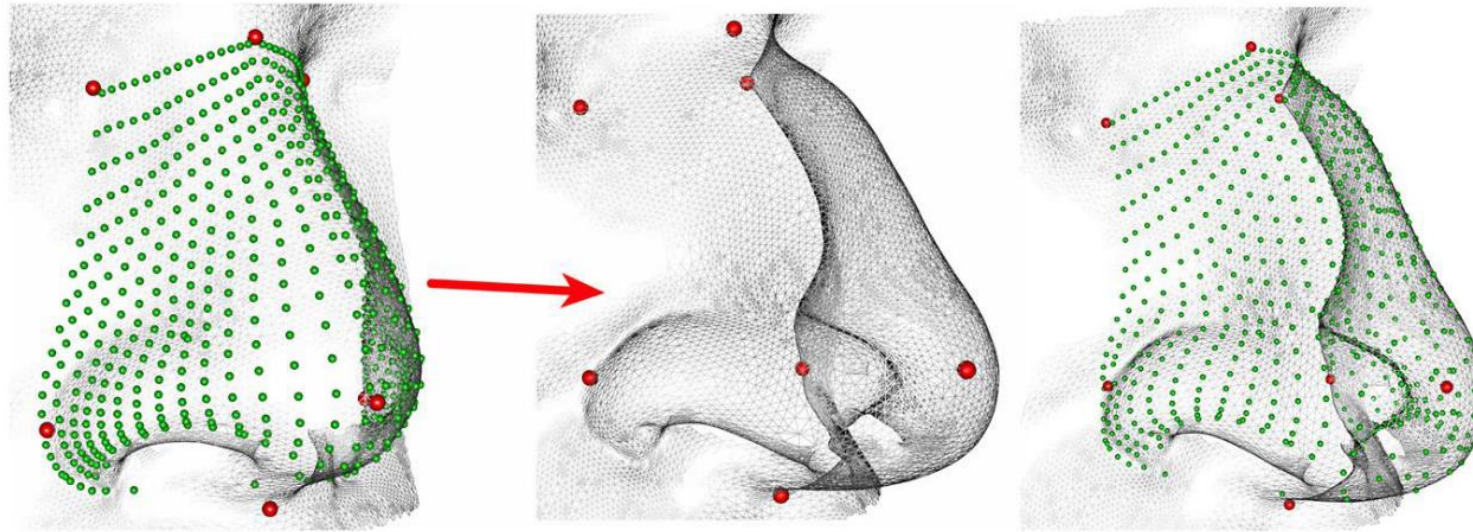
# The Problem



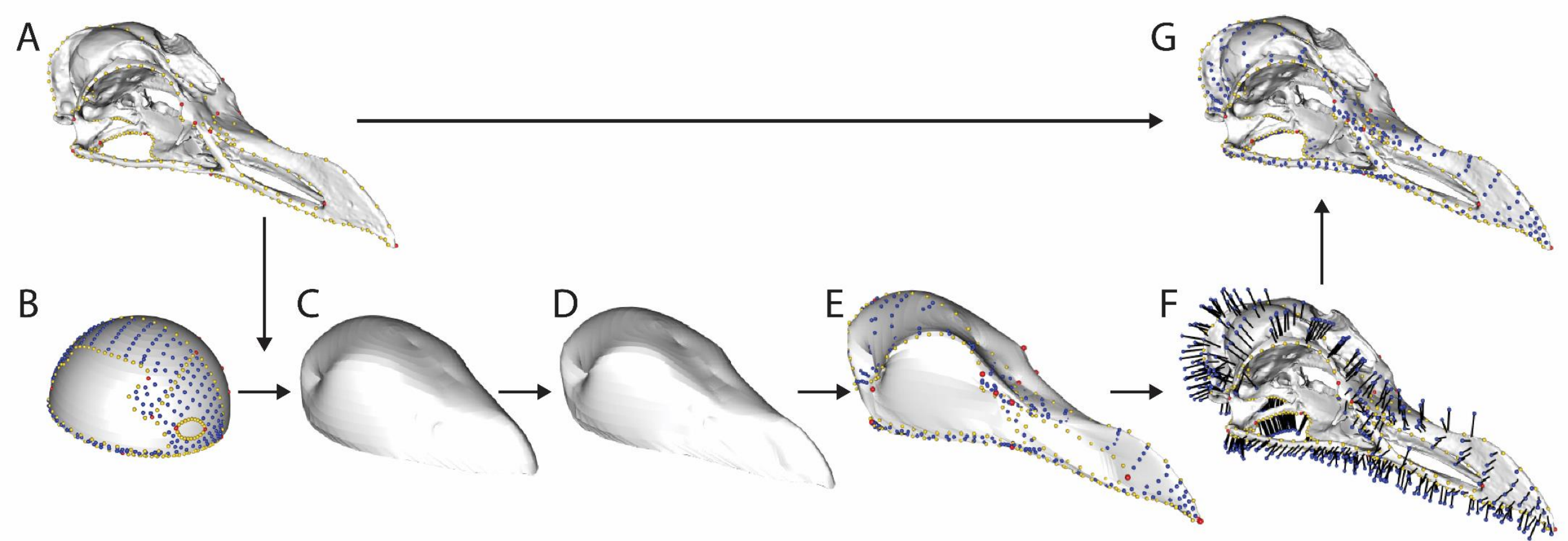


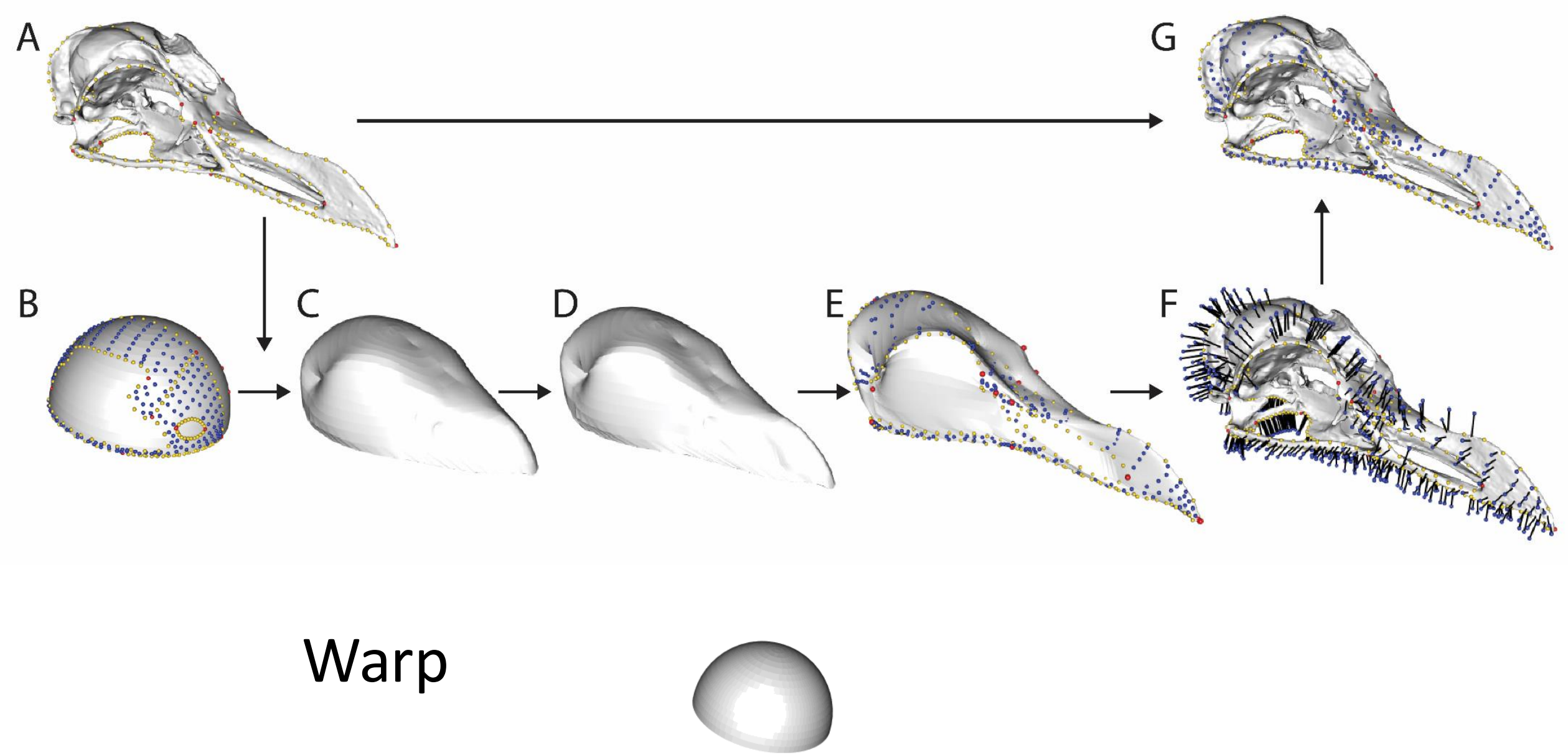


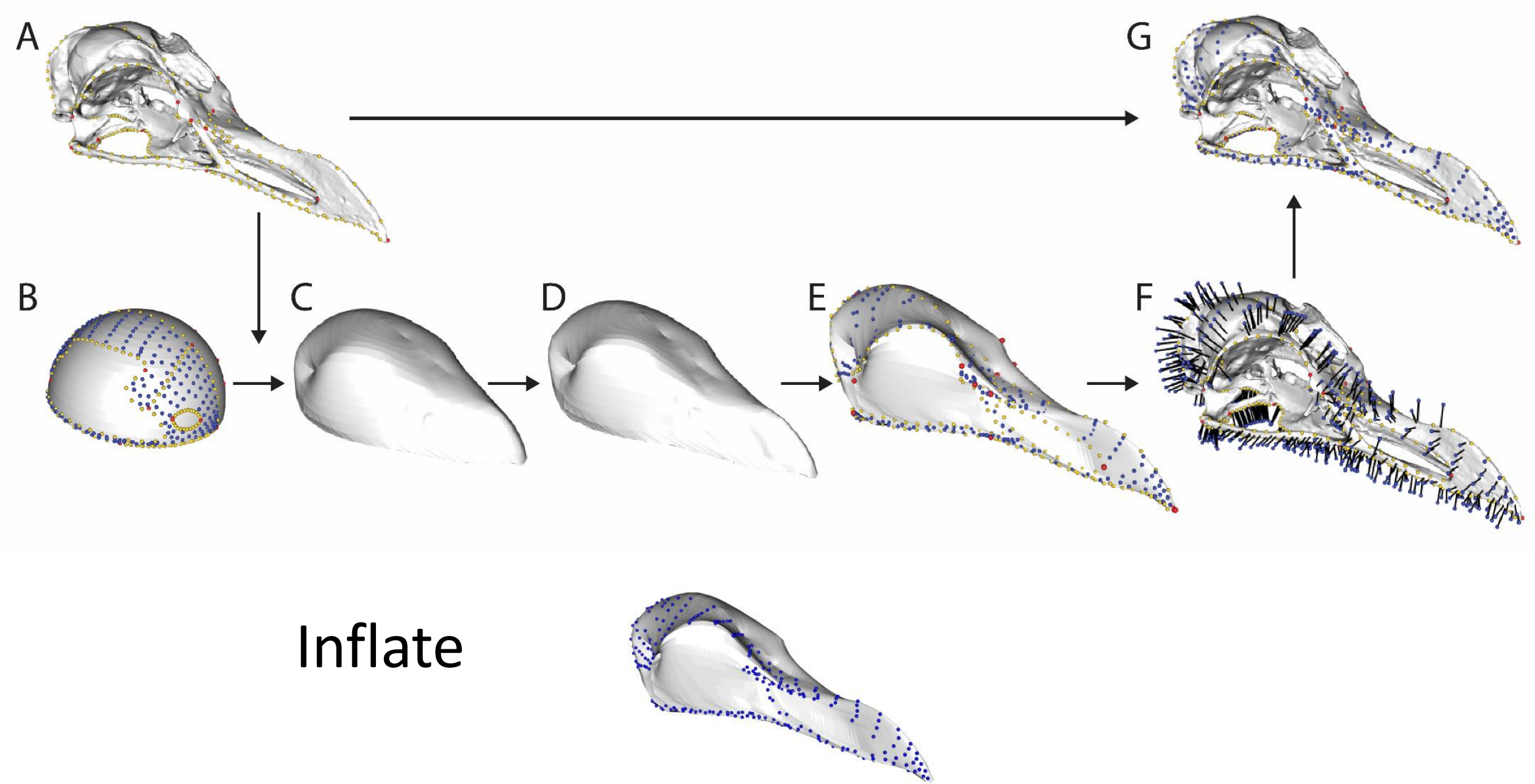
# Semi-automated patching

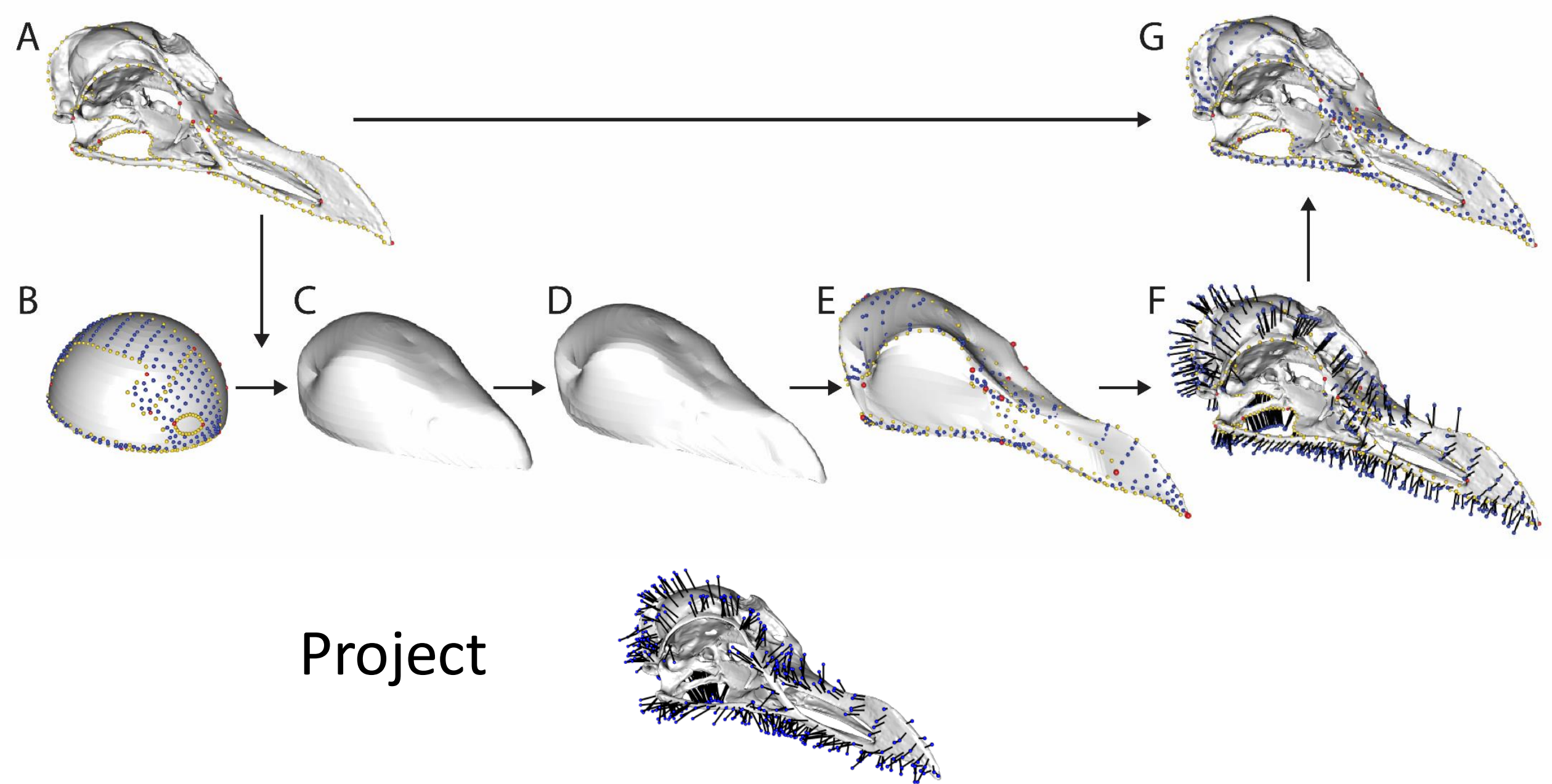


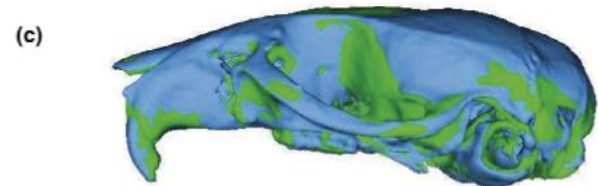
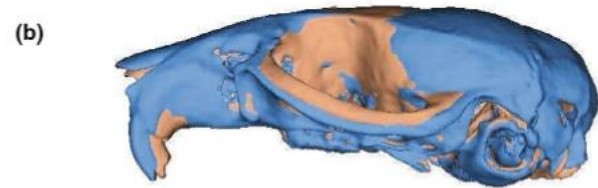
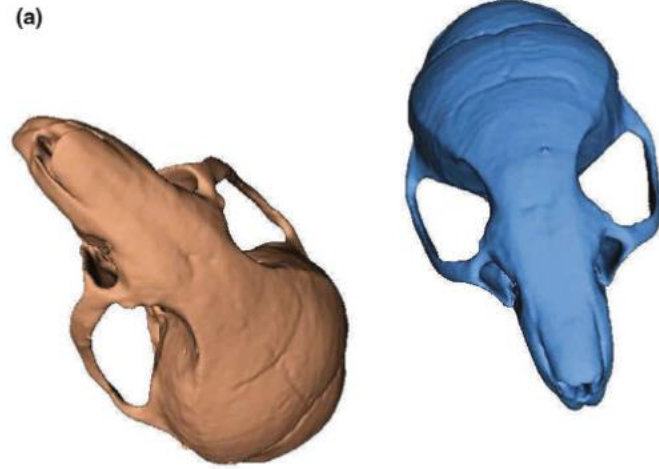
Morpho R package











Accepted: 11 July 2021

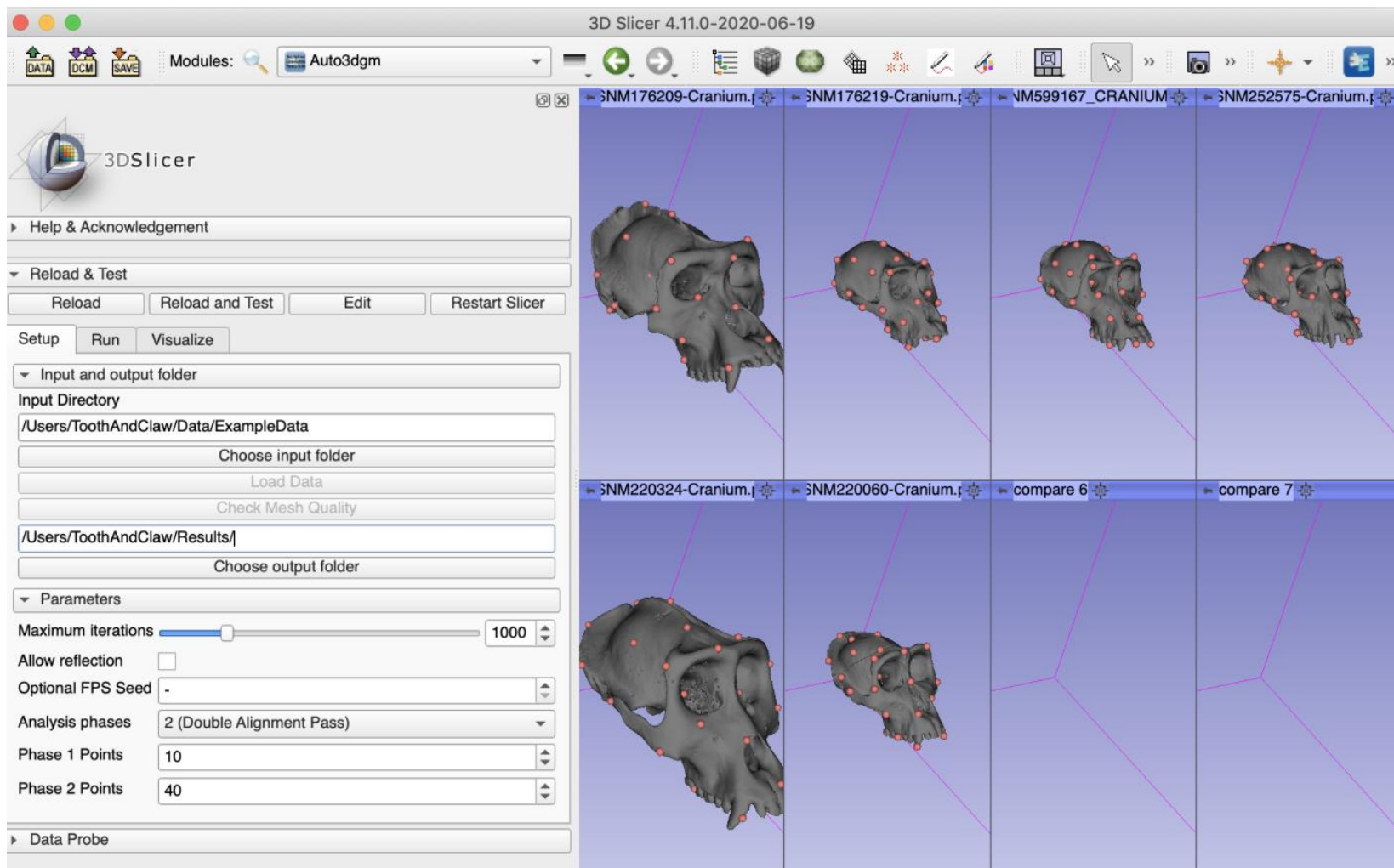
DOI: 10.1111/2041-210X.13689

## RESEARCH ARTICLE

Methods in Ecology and Evolution 

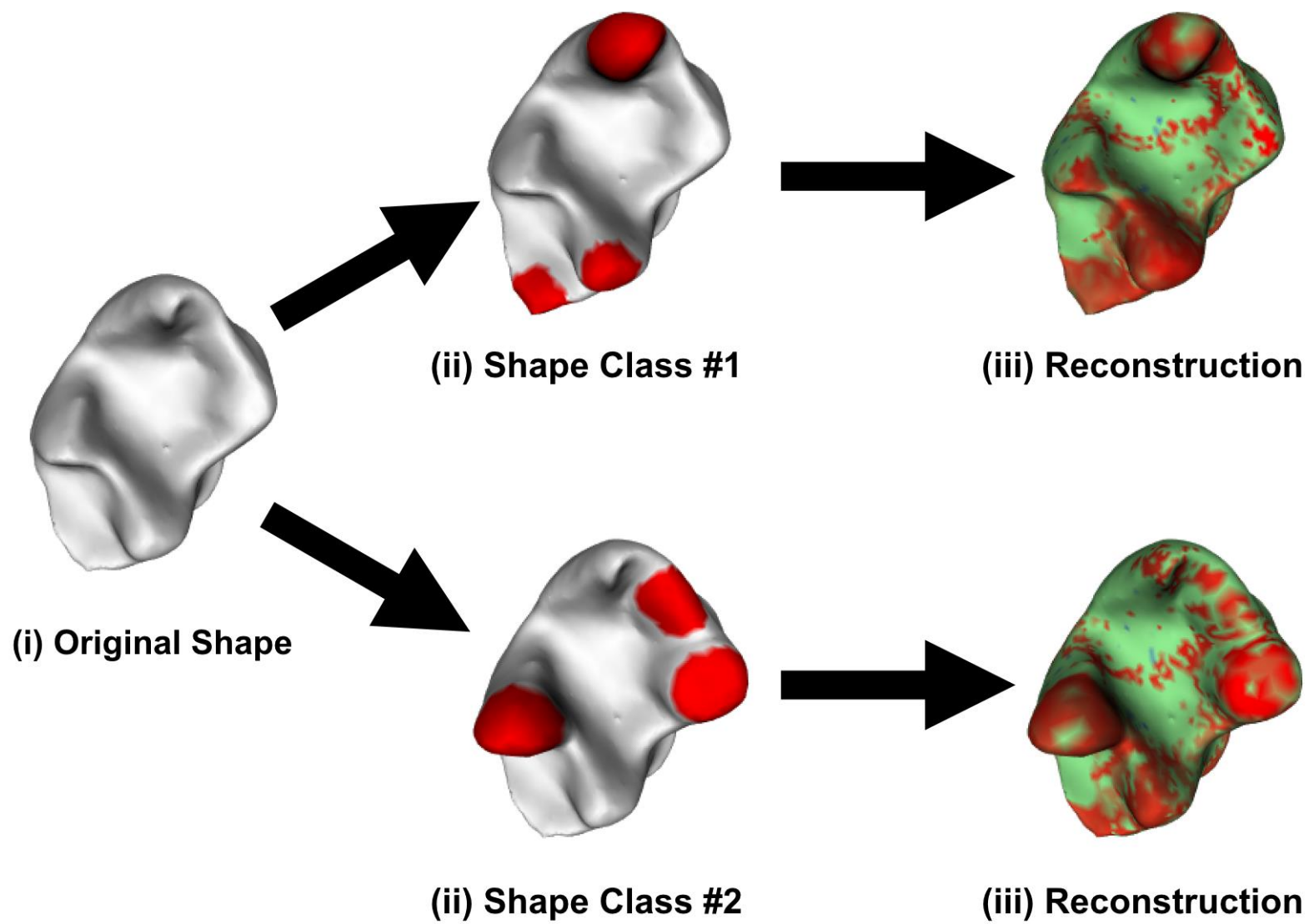
# ALPACA: A fast and accurate computer vision approach for automated landmarking of three-dimensional biological structures

Arthur Porto<sup>1,2</sup>  | Sara Rolfe<sup>3,4</sup> | A. Murat Maga<sup>4,5</sup> 



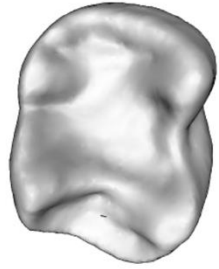
Pseudolandmarks-  
Auto3dGm

<https://toothandclaw.github.io>



### (a) Input 3D Shapes

#### Data from Species #1



Outcome:  $y_i = 0$

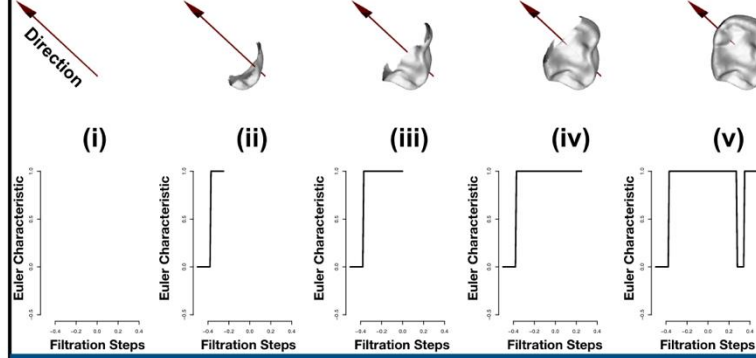
#### Data from Species #2



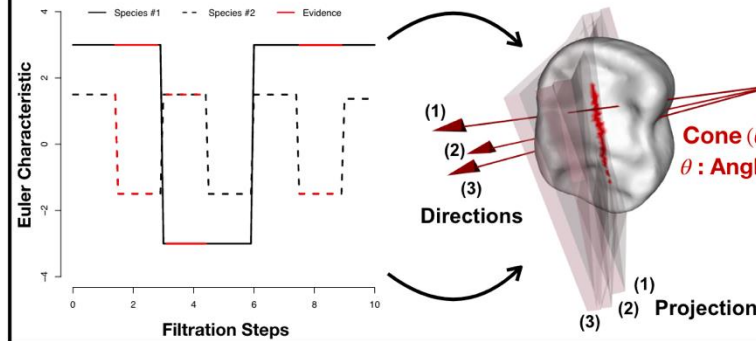
Outcome:  $y_i = 1$

### (b) Derive Topological Summary Statistics

#### Steps in Filtration Process:

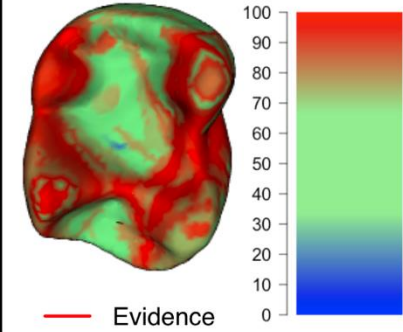


### (c) Variable Selection and Reconstruction

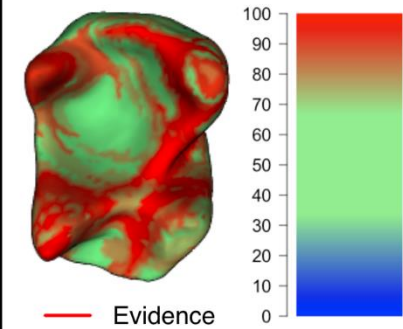


### (d) Visualize Enrichment

#### Enrichment in Species #1



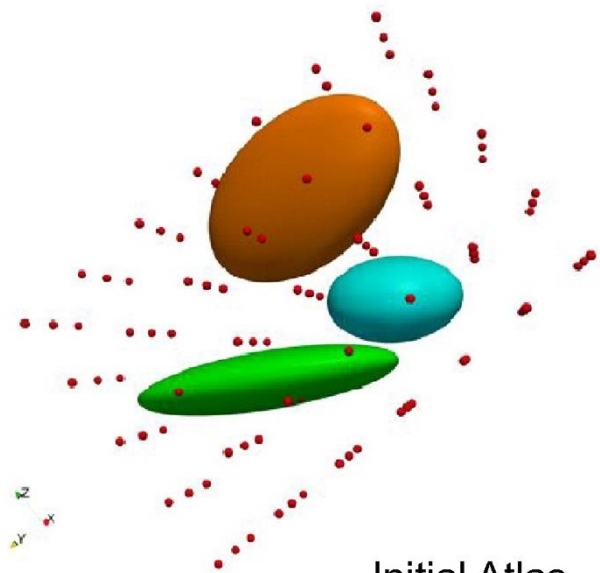
#### Enrichment in Species #2



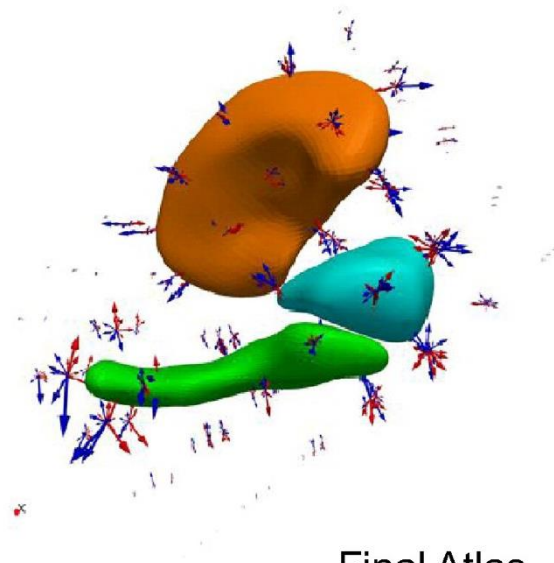
<sup>1</sup> **A Statistical Pipeline for Identifying Physical Features that**  
<sup>2</sup> **Differentiate Classes of 3D Shapes**

<sup>3</sup>  
<sup>4</sup> Bruce Wang<sup>1,2,\*</sup>, Timothy Sudijono<sup>3,\*</sup>, Henry Kirveslahti<sup>4,\*</sup>, Tingran Gao<sup>5</sup>, Douglas M. Boyer<sup>6</sup>, Sayan  
<sup>5</sup> Mukherjee<sup>4,7-9,†</sup>, and Lorin Crawford<sup>10,†</sup>

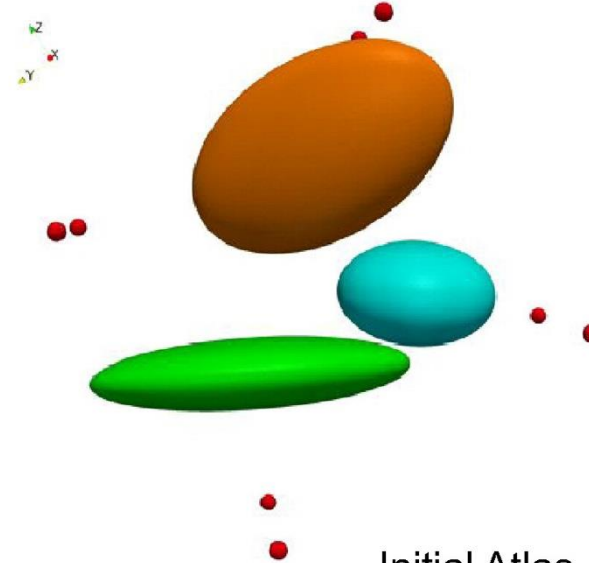
# Deterministic Atlas Analysis



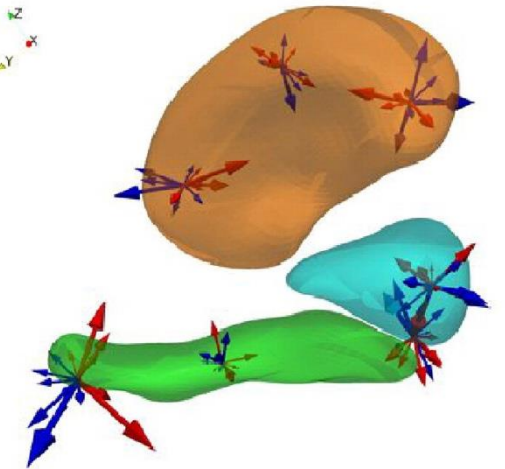
Initial Atlas



Final Atlas

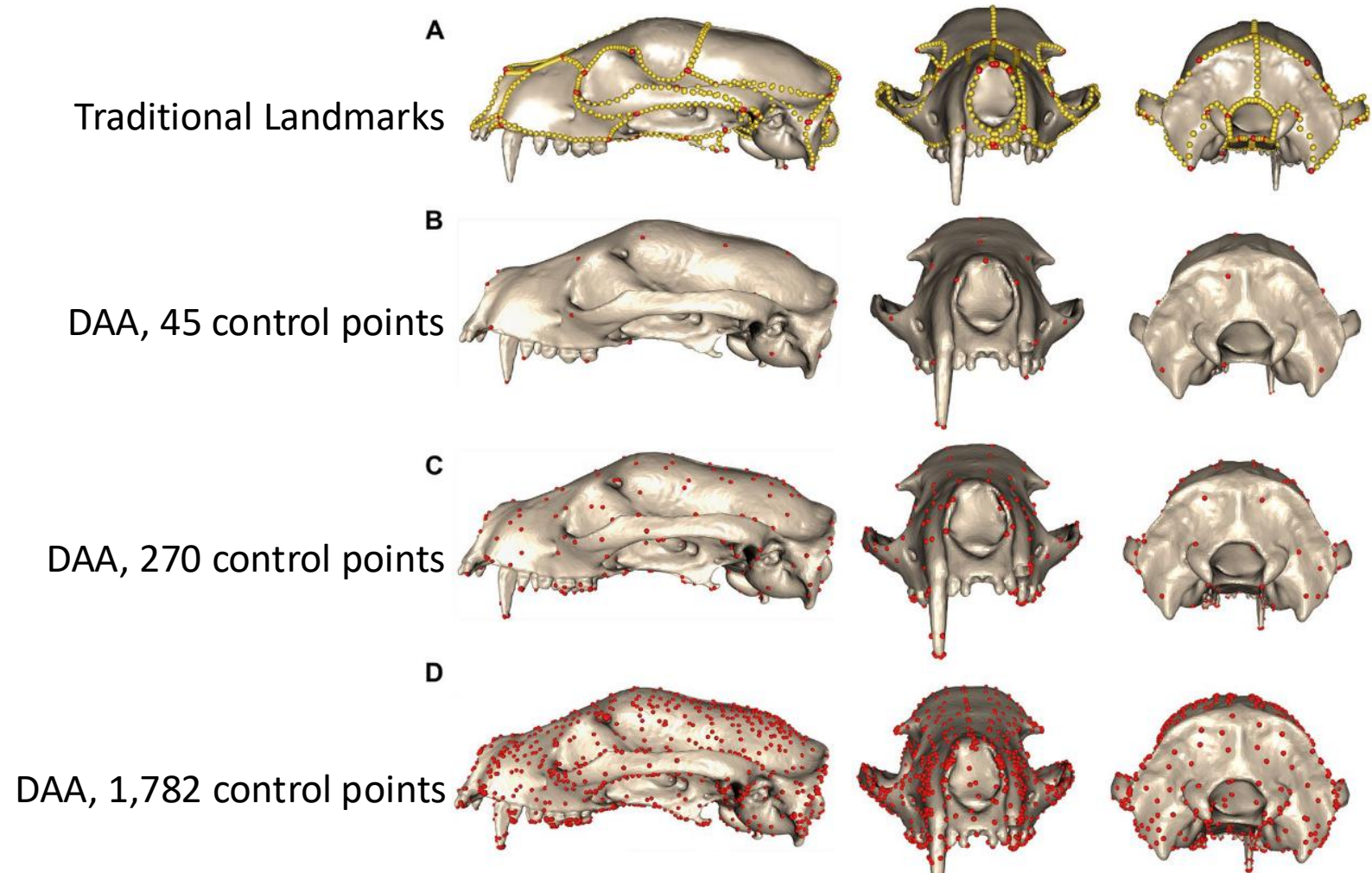


Initial Atlas



Final Atlas

# Deterministic Atlas Analysis



RESEARCH

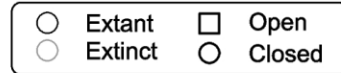
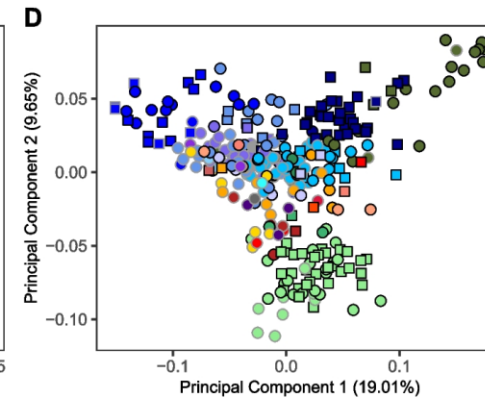
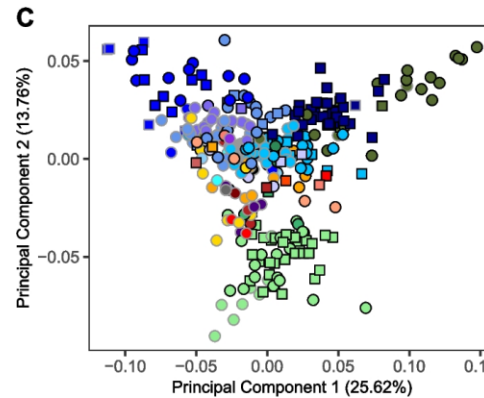
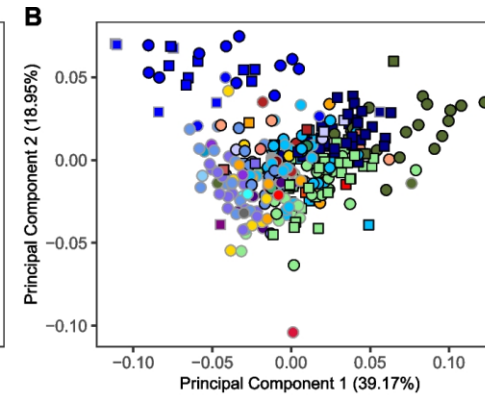
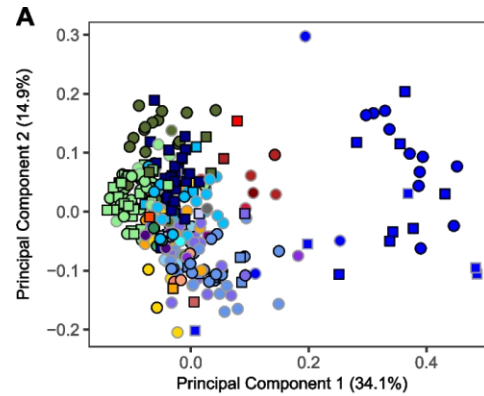
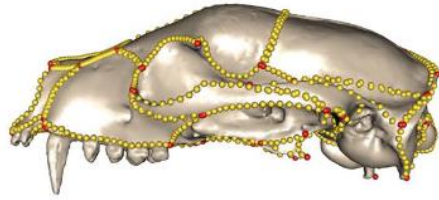
Open Access

Assessing the application of landmark-free morphometrics to macroevolutionary analyses

James M. Mulqueeney<sup>1,2\*</sup>, Thomas H. G. Ezard<sup>1</sup> and Anjali Goswami<sup>2</sup>



# Deterministic Atlas Analysis



## Stem

- Zalambdalestidae
- Cimolestia
- Leptictida

## Euarchontaglires

- Dermoptera
- Scandentia
- Primates
- Rodentia
- Lagomorpha

## Xenarthra

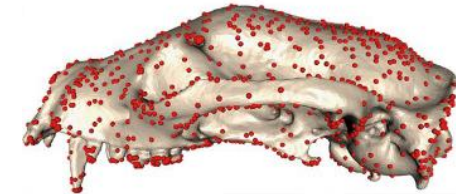
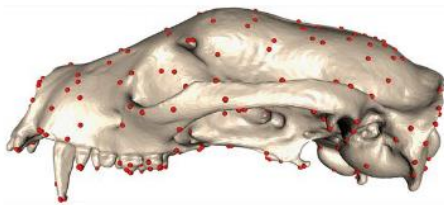
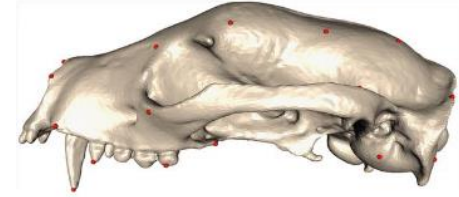
- Cingulata
- Pilosa

## Afrotheria

- Afrosoricida
- Macroscelidea
- Tubulidentata
- Hyracoidea
- Proboscidea
- Desmostylia
- Sirenia
- Embrithopoda

## Laurasiatheria

- Chiroptera
- Eulipotyphla
- Acreodi
- Amblypoda
- Artiodactyla
- Cetacea
- Litopterna
- Notoungulata
- Astrapotheria
- Perissodactyla
- Creodonta
- Carnivora
- Pholidota



# How much firepower do I need?



VS.



# Modelling High Dimensional Data

- What do we mean by high dimensional data?
- $N$  = number of observations
- $P$  = number of variables

|           | Trait 1 | Trait 2 | Trait 3 |
|-----------|---------|---------|---------|
| Species 1 |         |         |         |
| Species 2 |         |         |         |
| Species 3 |         |         |         |
| Species 4 |         |         |         |
| Species 5 |         |         |         |
| Species 6 |         |         |         |
| Species 7 |         |         |         |

|           | Trait 1 | Trait 2 | Trait 3 | Trait 4 | Trait 5 | Trait 6 |
|-----------|---------|---------|---------|---------|---------|---------|
| Species 1 |         |         |         |         |         |         |
| Species 2 |         |         |         |         |         |         |
| Species 3 |         |         |         |         |         |         |
| Species 4 |         |         |         |         |         |         |

# Modelling High Dimensional Data

- Who cares?
- Everything is linear models
- Estimating F-statistic and likelihood of linear models with  $p > N$  is impossible (you can't invert a matrix that has  $p > N$ )
- “Curse of dimensionality”

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \sigma$$

$$F = \frac{\hat{\boldsymbol{\beta}}^\top (\mathbf{X}^\top \mathbf{X}) \hat{\boldsymbol{\beta}} / p}{\hat{\sigma}^2}$$

# The Solution:

- RRPP: Residual randomization in permutation procedure
- F-statistic (and significance) is computed by randomizing the residuals of the model and comparing to a null

$$F = \frac{MS_{Model}}{MS_{residuals}}$$

MS = Mean squares

# The Solution:

- RRPP: Residual randomization in permutation procedure
- F-statistic (and significance) is computed by randomizing the residuals of the model and comparing to a null

## Recommendations:

- 1) Always think about what methods are appropriate for your data
- 2) Use `procD.lm` or `procD.pgls` functions in `geomorph` or `lm.rrpp` in the RRPP package

### APPLICATION

Methods in Ecology and Evolution



**RRPP: An R package for fitting linear models to high-dimensional data using residual randomization**

Michael L. Collyer<sup>1</sup>  | Dean C. Adams<sup>2,3</sup>

<https://doi.org/10.1111/2041-210X.13029>

